# Leakage-Free, Monotonic, and Uncertainty-Calibrated Bearing RUL Prediction for Reliability-Centered Maintenance

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**Abstract**

Rolling-element bearing remaining useful life (RUL) models may achieve low prediction error yet remain unreliable for maintenance when model selection uses test bearings, trajectories are non-monotonic, or uncertainty intervals are uncalibrated. This study presents a leakage-free, monotonicity-aware, uncertainty-calibrated framework instantiated by DSAFLite+PGMC, a 0.19 M-parameter dual-stream network adaptively fusing raw-vibration and Hilbert-envelope-spectrum features. Training budgets are selected exclusively from training bearings using two-fold bearing-level validation and fixed before held-out evaluation. A soft monotonic regularizer guides learning; PAVA provides retrospective smoothing, RTRM and CES provide causal correction, and MC-dropout intervals are calibrated without test-data access. On PRONOSTIA, across 20 seeds, raw MAE/$R^2$ reached $0.0897 \pm 0.0050/0.8169 \pm 0.0176$ and PAVA-smoothed MAE/$R^2$ reached $0.0789 \pm 0.0068/0.8646 \pm 0.0224$, outperforming five baselines. Ablations identify adaptive fusion and dual-stream learning as the main accuracy drivers, with monotonic and adversarial regularization as auxiliary priors. XJTU-SY confirms in-domain utility but limited zero-shot transfer ($R^2 = 0.19$). Calibration improved one-sided lower-bound coverage from 59.9% to 89.1% at 90% nominal coverage and reduced missed interval-trigger detections. Decision analysis shows that PAVA improves error metrics but may delay alarms under asymmetric late-maintenance costs. Reliable deployment therefore requires joint evaluation of model selection, monotonicity, uncertainty calibration, and maintenance outcomes.

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## Nomenclature

| Symbol | Definition |
|---|---|
| $x_t$ | Raw vibration window at step t; $x_t \in \mathbb{R}^{2560}$ (100 ms at 25.6 kHz) |
| $y_t$ | Normalized RUL label; $y_t$ = (T − t)/T ∈ [0,1] |
| $T$ | Total observation windows in a bearing's run-to-failure trajectory |
| $\mu_t, s_t$ | MC-dropout predictive mean and standard deviation at step t |
| $F_t^{temp}$ | Temporal feature vector, temporal 1D CNN output; $\in \mathbb{R}^{128}$ |
| $F_t^{spec}$ | Spectral feature vector, envelope-spectrum 1D CNN output; $\in \mathbb{R}^{128}$ |
| $\alpha_t$ | Adaptive scalar fusion gate weight; $\alpha_t \in [0,1]$ |
| $L_{PGMC}$ | Physics-Guided Monotonic Consistency loss; soft squared-hinge penalty on positive RUL increments over chronological training chunks, not a hard constraint |
| DANN | Domain-adversarial neural network; here, a training-only source-condition adversarial regularizer discarded at inference |
| $L_{DANN}$ | Binary cross-entropy source-condition classification loss through the gradient-reversal branch |
| $N_c$ | PGMC chunk length (chronological windows per penalty evaluation); $N_c$ = 32 |
| $c*$ | Post-hoc interval scale factor fitted on the two bearing-level validation folds (Section 4.3); c* = 2.41 |
| $z$ | Conventional two-sided 90% Gaussian half-width; z = 1.645 (uncalibrated starting scale, refined by c*) |
| $\tau$ | Maintenance trigger threshold (normalized RUL) |
| LBC | Lower-bound coverage, $P(y_t \geq \mu_t - zs_t)$ (Section 3.10); a one-sided coverage metric, distinct from the conventional two-sided interval-membership rate denoted PICP |
| PAVA | Pool-adjacent-violators algorithm (isotonic regression post-processing) |
| $B*$ | Selected training budget (epochs), chosen by two-fold bearing-level cross-validation (Section 4.3); B* = 5 for both PRONOSTIA and XJTU-SY |

## 1. Introduction

Rolling element bearings are among the most failure-prone components of rotating machinery, and unplanned bearing failure causes downtime and substantial economic loss in manufacturing, energy, and transportation [1, 2]. Vibration is the dominant sensing modality for bearing condition monitoring, and prognostics and health management (PHM) systems that deliver reliable remaining useful life (RUL) estimates enable condition-based maintenance [3], the operational objective that reliability-centered maintenance frameworks formalize [32]. RUL prediction is intrinsically harder than fault diagnosis: rather than classifying a health state, it must track a continuous degradation trajectory from early-stage operation to end-of-life using a vibration representation that stays informative across the full range of degradation severity. Critically, RUL prediction is not only a regression problem; it is a reliability decision-support problem, because a maintenance planner acts on the model's output, not on its point-error metric alone.

Three threats routinely separate a low reported error from a trustworthy maintenance decision. First, model-selection leakage: deep learning now dominates data-driven bearing RUL prediction because neural networks extract degradation-relevant features directly from raw sensor data [4, 5], but run-to-failure benchmarks contain few test units (four held-out bearings in the standard PRONOSTIA split), and the model-selection step, not only the final metric, is a major source of optimism. On small run-to-failure datasets, test-bearing reuse during checkpoint or hyperparameter selection can change which representation or architecture appears superior, which is especially problematic when comparing raw, spectral, recurrent, and transformer-based models. We show below that early stopping monitored on a validation split drawn from the training bearings selects checkpoints that overfit those bearings: on PRONOSTIA, checkpoints selected this way degrade held-out-bearing MAE by 41% relative to the budget chosen by our two-fold bearing-level validation criterion. Concerns of this type are well documented in the broader model-selection literature [9, 10] but are rarely quantified for PHM benchmarks. Second, non-monotonic degradation trajectories: RUL should generally decrease over a run-to-failure trajectory, yet networks trained by pointwise regression losses routinely produce non-monotonic predictions: in our experiments, raw network outputs are monotonically non-increasing for only ≈51% of consecutive prediction pairs, indistinguishable from chance. Training-time monotonicity penalties and post-hoc isotonic correction [11] address different parts of this problem, and the causal (real-time) analogues of the latter have received little systematic analysis in the PHM literature. Third, uncalibrated predictive uncertainty: maintenance decisions require calibrated confidence, not only point estimates [12, 13]; MC-dropout is known to produce intervals that are too narrow [14], and few bearing RUL papers verify coverage before using an interval to trigger a decision rule. Each threat can independently produce an unsafe or economically poor maintenance decision even when the reported point-accuracy metric looks strong, and without controlling all three, claims that one representation, fusion strategy, or uncertainty method outperforms another cannot be trusted.

Two complementary vibration representations recur in this literature and serve here as degradation-evidence channels rather than the paper's central novelty. Raw time-domain vibration preserves phase, impulsiveness, and the non-stationary transient structure produced by an evolving defect, and convolutional networks applied directly to raw windows consistently outperform hand-crafted features for accelerated degradation data [4, 5]. The Hilbert-envelope spectrum instead demodulates the signal to expose amplitude modulation at the bearing's characteristic defect frequencies, a representation with a long history in bearing diagnostics [6]. Most deep-learning RUL models commit to a single representation, raw vibration [7] or a time-frequency image [8], which discards the complementary evidence available in the other domain: raw-only models can be noise-sensitive, while spectral-only models can lose the time-domain degradation cues embedded in fault-impulse morphology. A dual representation that retains both is therefore physically, not only statistically, motivated.

Even when both representations are available, their relative importance is unlikely to stay constant across a bearing's life: early-stage signals may be dominated by broadband raw-vibration texture, while an emergent localized defect may first become visible as narrowband modulation in the envelope spectrum, and either cue can be more or less reliable depending on operating noise. Fixed-weight fusion (simple concatenation or an unweighted average) cannot express this time-varying reliability. An adaptive, low-parameter gate that learns per-window how much to trust each stream offers an interpretable alternative, and its response can itself be inspected as a diagnostic, for example under injected noise, rather than treated as a black box. Because neither the gate nor a training-time monotonic penalty can themselves guarantee a deployable trajectory, we separate training-time regularization from evaluation-time monotonicity correction, and further separate the latter into an offline method used only for retrospective analysis and causal methods valid for online deployment.

Realizing any of these contributions requires an evaluation protocol capable of detecting it reliably. All claims below are therefore obtained under a leakage-free bearing-level protocol: the training budget is selected using two-fold validation over the training bearings only, the final configuration is then fixed and retrained, and held-out bearings are evaluated exactly once. Predictive intervals are likewise calibrated using only validation-fold residuals, frozen before touching the test bearings. This ordering (protocol before architecture) governs how the rest of the paper should be read: the dual-stream network described below is the vehicle used to demonstrate the protocol, not a claim that stands independently of it.

This paper makes five contributions to reliability-oriented bearing prognostics. First, a leakage-free model-selection protocol: training-budget selection is performed using only training bearings through two-fold bearing-level validation, followed by retraining and one-time evaluation on held-out bearings, preventing test-bearing information from influencing architecture or checkpoint choice. Second, a monotonic degradation-consistency analysis that separates a soft training-time regularizer (PGMC) from offline (PAVA) and causal (RTRM, CES) trajectory correction, clarifying which combination is actually deployable online. Third, calibrated predictive uncertainty for maintenance support: MC-dropout intervals are calibrated leak-free on validation-fold residuals and evaluated by coverage, not reported as an uncalibrated by-product. Fourth, a maintenance-threshold decision analysis that reframes the interval-triggered maintenance rule in terms of alarm timing and asymmetric decision loss rather than point classification alone, showing that the estimator with the best point accuracy is not always the one with the best decision outcome. Fifth, a compact dual-evidence prognostic estimator, DSAFLite+PGMC, that instantiates the framework with a 0.19 M-parameter network adaptively fusing raw-vibration temporal evidence and Hilbert-envelope spectral evidence, evaluated for noise robustness, cross-dataset transfer limits, and computational cost so that accuracy, reliability, and deployment constraints can be interpreted together.

None of the individual components above (1D convolutional streams, envelope-spectrum demodulation, adaptive gating, domain-adversarial regularization, isotonic post-processing, or MC-dropout uncertainty) is novel in isolation. The novelty is not a single isolated module. The contribution of this work is a reliability-oriented integration and evaluation protocol that connects representation learning, leakage-free model selection, monotonic degradation behavior, calibrated uncertainty, and maintenance decision support: demonstrating when and why raw-vibration and envelope-spectrum evidence should be adaptively combined rather than fixed or single-stream, showing that model-selection leakage can change which architectural conclusions appear to hold, and showing that the estimator favored by point-accuracy metrics is not necessarily the one favored by a downstream maintenance-decision rule.

We validate the framework on two run-to-failure datasets, PRONOSTIA [15] and XJTU-SY [16], under an identical protocol, including zero-shot transfer to an entirely unseen operating condition. We additionally report a negative result: naive cross-dataset transfer (PRONOSTIA → CWRU [17]) fails ($R^2 < 0$), and we explain why pseudo-RUL constructions over seeded-fault data do not support zero-shot claims.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 details the reliability-oriented prognostic framework instantiated by the proposed dual-stream network. Section 4 describes the leakage-free evaluation protocol and benchmark design. Section 5 reports results. Section 6 discusses findings and limitations, including a maintenance-threshold alarm-timing and decision-loss analysis that extends the interval-triggered maintenance rule. Section 7 concludes.

## 2. Related Work

### 2.1. Deep learning for bearing RUL prediction

CNN architectures applied to raw vibration consistently outperform hand-crafted features for PRONOSTIA-style accelerated degradation [7]. Recurrent hybrids add temporal context at higher parameter cost [20, 21], and transformer/MLP-mixer variants [22] target long-range dependencies but exhibit high training variance on small bearing datasets. None of these single-stream approaches incorporates physics-guided constraints or adaptive multi-modal fusion, and few report calibrated uncertainty [23]. Reliability-oriented evaluation requirements (leakage-free model selection and calibrated interval coverage) have not been systematically characterized in the deep-learning bearing-RUL literature.

### 2.2. Model-selection leakage and evaluation validity in PHM

Selection-induced optimism is a classical hazard in machine learning [9]: hyperparameters or checkpoints tuned on data that informs the final evaluation inflate reported performance, and the cross-validation guidance developed to control it [10] is rarely applied end-to-end in deep bearing-RUL pipelines. PHM evaluation guidance [30, 33] predates the deep-learning era's heavy checkpoint selection and seed variance, and vibration-based bearing diagnostics specifically has been shown to carry the same leakage risk in practice, with a large majority of prior CWRU-based studies found to use leakage-prone window splits [44]. With as few as four held-out bearings in the standard PRONOSTIA split, validation-split reuse during checkpoint or hyperparameter selection can change which architecture or fusion strategy appears superior, and test-bearing reuse specifically invalidates any downstream maintenance-risk claim built on the reported metric. The reliability consequence (undeclared risk transferred to the operator when an inflated metric informs a safety margin) has not been previously quantified on standard bearing PHM benchmarks; Section 4.2 quantifies it directly.

### 2.3. Monotonicity in RUL prediction

The physical constraint that RUL is non-increasing is not satisfied by standard training. Isotonic regression post-processing [11] guarantees monotonicity retrospectively but requires the complete sequence, which makes it valid for retrospective fleet analysis but not for streaming prognosis; training-time regularization penalties reduce violations during training but do not themselves guarantee ordering. We combine both, analyze the interaction, and add causal filters for real-time use. Distinctly from prior work, we show the regularizer's warm-up must be budget-scaled: with absolute warm-up schedules and a rigorously selected (short) budget, the regularizer never activates: a subtle failure mode that invalidates ablation conclusions if unnoticed.

### 2.4. Calibrated uncertainty and maintenance thresholds

Point RUL estimates are insufficient for maintenance decisions, which require calibrated confidence rather than a single number [12, 13]. MC-dropout [14] and broader UQ practice in prognostics [23] motivate interval reporting with explicit calibration metrics, but neural-network uncertainty estimates are known to undercover without post-hoc calibration [37, 38], and few bearing-RUL studies verify coverage before an interval is used to trigger a maintenance rule, let alone examine whether the resulting alarm arrives at a decision-relevant time. Recent RESS studies benchmark alternative UQ approaches for prognostics, including deep ensembles [39] and multi-distribution fusion [34, 35, 36], and conformal prediction offers a distribution-free alternative to variance-scaling calibration for RUL intervals specifically [40, 41], though it is not evaluated here (Section 6.5). Threshold-triggered maintenance policies built on imperfect RUL prognostics have themselves been shown to trade missed and false alarms in ways that depend on the underlying interval quality [42, 43]. The reliability consequence (incorrect coverage silently miscalibrating a maintenance-threshold decision) has not been previously connected to alarm timing or asymmetric decision loss on standard bearing PHM benchmarks; Section 6.3 extends coverage verification into exactly this decision-timing analysis.

### 2.5. Vibration representations and multi-modal fusion for bearing prognostics

Time-frequency images enable 2D backbones [8] at substantial computational cost. Subdomain-alignment networks such as HIWSAN [18] and SC-SAN [19] achieve strong PRONOSTIA transfer-task results (per-task MAE 0.071–0.091) using multi-scale feature alignment, but rely on static fusion and require 2.6 M parameters with $\approx$11 ms CPU latency [18]. Our adaptive gate performs per-window stream weighting at negligible cost and, in contrast to earlier reports of null fusion-gate effects, we demonstrate a consistent accuracy contribution under a leakage-free protocol and a directional gate response to noise. The envelope spectrum used in our spectral stream is grounded in this classical amplitude-demodulation analysis of rolling-element bearing faults [6], rather than being an arbitrary spectral transform. These representations function here as evidence channels feeding the reliability-oriented framework above, not as the paper's central claim.

### 2.6. Domain adaptation and cross-condition generalization

Domain-adversarial training [24] suppresses condition-specific information via gradient reversal, and has been applied to bearing diagnosis [25, 26] and prognosis [27, 28]. Alignment baselines include CORAL [29]. We use a lightweight DANN-style source-condition adversarial regularizer, not a full domain-adaptation module, over the operating conditions present in the training set only (no test or target-condition data of any kind participates in adversarial training) and assess its value via zero-shot extrapolation to an unseen XJTU-SY condition, which should be read as a test of extrapolation rather than of target-adapted transfer.

## 3. Reliability-Oriented Prognostic Framework

The overall reliability-oriented prognostic workflow is summarized in Figure 1.

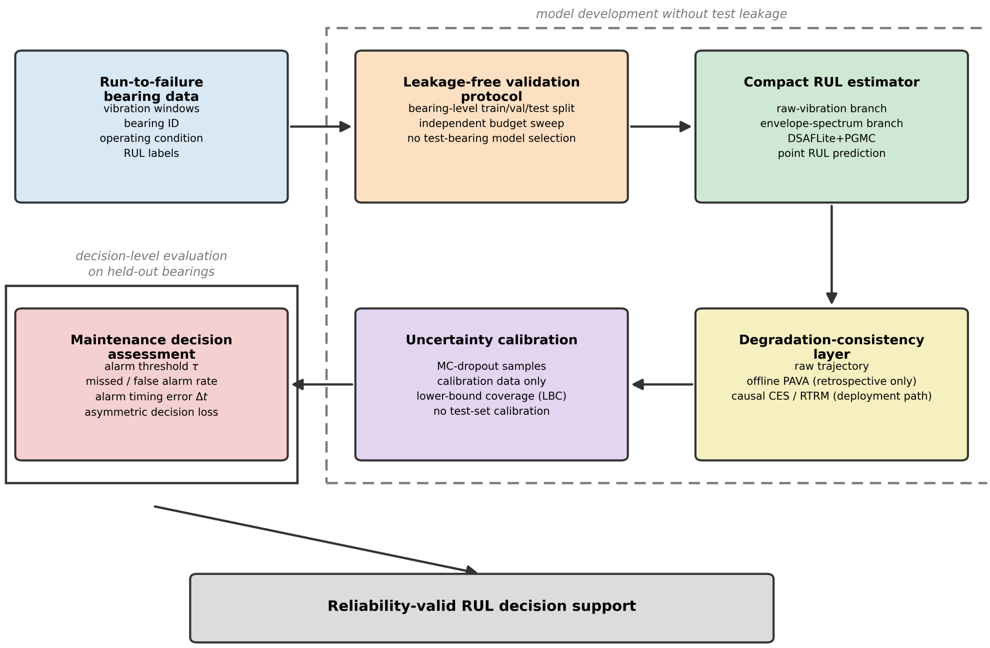


Figure 1: Reliability-oriented workflow for bearing RUL prediction and maintenance decision support. The workflow separates leakage-free bearing-level model selection from final testing, distinguishes retrospective monotonic smoothing (offline PAVA) from deployable causal filtering (RTRM, CES), calibrates uncertainty without test-set information, and evaluates predictions using both error metrics and maintenance-threshold decision metrics. The dashed boundary marks the stages of model development that never touch the held-out test bearings; the solid boundary marks the decision-level evaluation performed on those bearings.

### *3.1. Problem formulation and reliability requirements*

Let a bearing's run-to-failure trajectory be a sequence of T observation windows. At step t ∈ 1, …, T, a raw vibration window $x_t \in \mathbb{R}^{2560}$ (100 ms at 25.6 kHz) is captured. The normalized RUL label is

$$y_t = \frac{T - t}{T} \in [0, 1], \tag{1}$$

strictly non-increasing by construction. The objective is a mapping f: $x_t \mapsto \hat{y}_t$ minimizing mean absolute error (MAE) on bearings never seen in training, while respecting monotonicity and providing calibrated uncertainty. Concretely, a reliability-valid RUL estimator should satisfy four conditions: (i) model selection must not access held-out bearings (Sections 4.2– 4.3); (ii) predicted trajectories should be degradation-consistent, or corrected with a declared offline or causal method (Section 3.9); (iii) predictive intervals should be calibrated before decision use (Section 3.10); and (iv) claims must be bounded to the operating regimes represented in the validation data (Section 6.5). The remainder of this section describes DSAFLite+PGMC, the compact dual-stream estimator used to instantiate this framework; it is the vehicle for demonstrating the four conditions above, not an independent claim.

### 3.2. Overall architecture

DSAFLite+PGMC comprises: (i) a temporal stream (1D CNN on raw vibration); (ii) a spectral stream (fixed Hilbert-envelope-spectrum extractor + 1D CNN); (iii) an adaptive scalar fusion gate; (iv) a regression head; (v) training-time losses (PGMC, DANN); and (vi) evaluation-time post-processing (per-bearing PAVA offline, or causal filters online). The DANN discriminator and PGMC penalty are discarded at inference; the inference graph totals 192,130 parameters. The complete DSAFLite+PGMC architecture is illustrated in Figure 2.

### 3.3. Temporal stream

The raw window $x_t$ (1×2560) passes through three 1D convolution blocks (kernel sizes 15/7/3, strides 4/4/2, channels 1→64→128→128, BatchNorm + GELU), global average pooling, and a linear projection to $F_t^{\text{temp}} \in \mathbb{R}^{128}$.

The spectral stream is based on the envelope spectrum rather than a generic Fourier transform. For bearing faults, localized defects excite resonances that are amplitude-modulated at characteristic defect frequencies and their harmonics; Hilbert-envelope demodulation suppresses carrier-frequency dependence and exposes this modulation structure [6]. The fixed envelope-spectrum front-end therefore injects a diagnostic prior while leaving the subsequent CNN to learn degradation-relevant spectral patterns from data.

### 3.4. Spectral stream (envelope spectrum)

The same $x_t$ is routed to a fixed, non-learnable envelope extractor. Given $x_t \in \mathbb{R}^N$ (N = 2560), the FFT-based analytic signal is:

$$\dot{x}_t = \mathrm{IFFT}(W \odot \mathrm{FFT}(x_t)), \qquad W[k] = \begin{cases} 2 & \text{if } 1 \le k < N/2 \\ 1 & \text{if } k = 0 \text{ or } N/2 \\ 0 & \text{if } k > N/2 \end{cases} \tag{2}$$

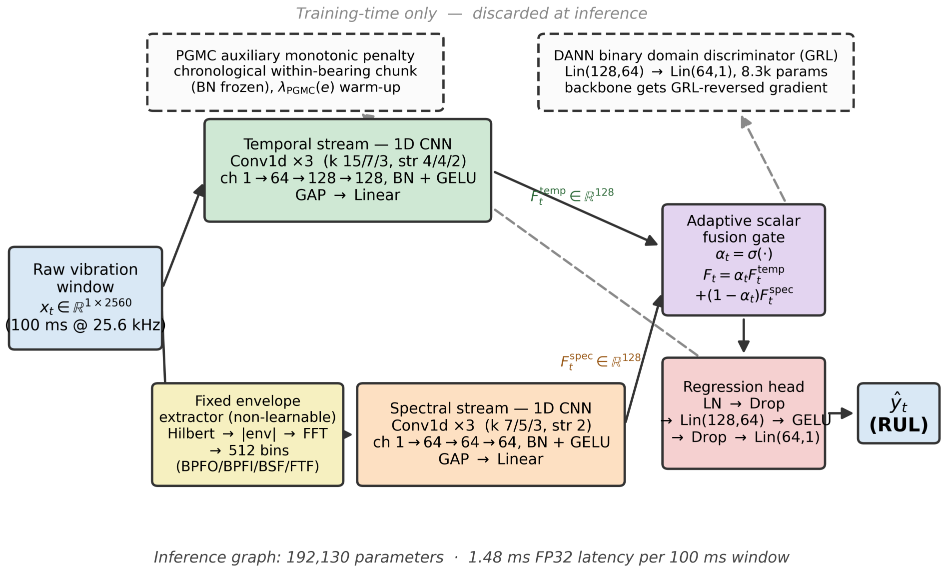


Figure 2: DSAFLite+PGMC architecture. Raw vibration drives a temporal 1D CNN and, in parallel, a fixed (non-learnable) Hilbert-envelope-spectrum extractor feeding a compact 3-layer 1D CNN; an adaptive scalar gate fuses the two 128-dimensional streams, and a regression head outputs RUL. The PGMC auxiliary monotonic penalty and the binary DANN domain discriminator (dashed) act only during training and are discarded at inference (192,130 parameters, 1.48 ms FP32 per 100 ms window).

where $\odot$ denotes element-wise multiplication. The envelope is $e_t[\mathrm{n}] = |\dot{x}_t[\mathrm{n}]|$, and the one-sided envelope spectrum is:

$$S_t[k] = |\mathrm{FFT}\{e_t\}[k]|, \qquad k = 0, 1, \ldots, N/2 \tag{3}$$

Normalized to unit maximum and truncated to M = 512 bins (covering 0–5,120 Hz at 10 Hz resolution). For a standard rolling-element bearing, the characteristic defect frequencies are BPFO = (n/2)$f_r$(1 − (d/D)cos$\theta$), BPFI = (n/2)$f_r$(1 + (d/D)cos$\theta$), BSF = (D/2d)$f_r$[1 − (d/D)$^2$cos$^2\theta$], and FTF = (1/2)$f_r$(1 − (d/D)cos$\theta$), where n is the number of rolling elements, d and D the ball and pitch diameters, $\theta$ the contact angle, and $f_r$ the shaft rotational frequency [6]. At the PRONOSTIA 6205 and XJTU-SY shaft speeds used here ($f_r \approx$ 27–40 Hz), these frequencies and their low harmonics remain well below 1,000 Hz for standard deep-groove ball-bearing geometries, so no defect-frequency clipping occurs. $S_t$ covers this defect-frequency band and feeds a 3-layer spectral 1D CNN (channels 1→64→64→64, kernels 7/5/3), global average pooling, and a linear projection to produce $F_t^{\mathrm{spec}} \in \mathbb{R}^{128}$.

### 3.5. Adaptive scalar fusion gate

A two-layer MLP over the concatenated features emits $\alpha_t \in [0,1]$:

$$\alpha_t = \sigma\left(W_2 \cdot \mathrm{GELU}\left(W_1 \left[F_t^{\mathrm{temp}}; F_t^{\mathrm{spec}}\right]\right)\right), \qquad F_t = \alpha_t F_t^{\mathrm{temp}} + (1 - \alpha_t) F_t^{\mathrm{spec}}. \tag{4}$$

The gate weights the temporal stream by its instantaneous reliability. Section 5.3 shows $\alpha_t$ responds to bearing identity (per-bearing means 0.33–0.47), and Section 5.5 shows a further response to injected noise (monotone shift toward the spectral stream with decreasing SNR).

### 3.6. Soft physics-guided monotonic consistency (PGMC)

PGMC is a soft, differentiable training regularizer (not a hard constraint on model outputs) that penalizes physically invalid positive increments of predicted RUL over a chronologically ordered sub-sequence of $N_c$ windows drawn from a single training bearing, where $N_c = 32$ is the auxiliary-chunk length, a contiguous sub-sequence of the full trajectory of length T defined in Section 3.1 ($N_c \ll$ T):

$$L_{\mathrm{PGMC}} = \frac{1}{N_c - 1} \sum_{t=1}^{N_c - 1} \max(0, \hat{y}_{t+1} - \hat{y}_t)^2, \tag{5}$$

where the chunk predictions $\hat{y}_1 \ldots \hat{y}_{N_c}$ are produced by an auxiliary forward pass with BatchNorm layers held in evaluation mode so that the pass does not perturb running statistics. The total loss is

$$L = L_{\mathrm{MAE}} + \lambda_{\mathrm{PGMC}}(e) \cdot L_{\mathrm{PGMC}} + \lambda_{\mathrm{DANN}} \cdot L_{\mathrm{DANN}}, \tag{6}$$

where e is the training-epoch index and the regularizer strength follows a linear warm-up $\lambda_{\mathrm{PGMC}}$(e) = $\lambda_{\max}$ · min(1, e / $w_{\mathrm{PGMC}}$), with $\lambda_{\max}$ = 0.08.

Budget-scaled warm-up. The warm-up length must scale with the training budget B: we use $w_{\mathrm{PGMC}}$ = max(2, ⌈B·10/120⌉). With an absolute warm-up designed for long schedules (e.g., 10 epochs) and a rigorously selected short budget (Section 4.3 selects B* = 5), the regularizer would otherwise never reach full strength and a DANN warm-up of 20 epochs would never activate at all. Our ablations (Section 5.3) quantify this effect.

Batch health-ordering variant. We also evaluate a relaxation that sorts each random mini-batch by descending target RUL and penalizes prediction increments along that ordering, a health-ordering consistency prior that requires no sequential batch construction. Both instantiations are compared in Section 5.3.

PGMC is a lightweight degradation-direction regularizer, not a hard step-level monotonicity guarantee: it shapes the latent trajectory so that post-hoc isotonic correction can recover a well-ordered sequence, while the hard monotonicity guarantee itself is supplied by the post-processing stage (Section 3.9).

### *3.7. Source-condition adversarial regularization (DANN)*

A gradient-reversal discriminator [24] is attached to the fused representation as a training-only regularizer, not as a standalone contribution. Domain labels are the operating conditions of the training bearings only, and in both datasets the training set spans exactly two operating conditions, so the domain-classification task is binary in every experiment reported here. For PRONOSTIA the two domains are Condition 1 (Bearing1_1, Bearing1_2) and Condition 2 (Bearing2_1, Bearing2_2); for XJTU-SY they are Condition 1 (Bearing1_1, Bearing1_2) and Condition 2 (Bearing2_1, Bearing2_2), with Condition 3 withheld entirely for zero-shot evaluation and therefore absent from adversarial training. The discriminator is a single binary head Linear (128→64) → GELU → Linear (64→1), 8,321 parameters, preceded by a gradient-reversal layer (GRL). Each training step makes two passes through this head: first, the GRL receives the fused feature directly, with no detachment, so the backbone parameters receive the sign-reversed gradient of the binary cross-entropy domain loss through this path ($\lambda_{\mathrm{DANN}}$ = 0.08 after a budget-scaled warm-up $w_{\mathrm{DANN}}$ = max(3, ⌈B·20/120⌉)); second, a fresh, detached copy of the fused feature (carrying no gradient to the backbone) is used to update only the discriminator's own parameters with a separate optimizer, a standard stabilization step that keeps the discriminator's own training signal undistorted by the sign reversal it is simultaneously subject to in the first pass. Every training batch mixes both conditions. No test or evaluation data participates in adversarial training in any form, and the discriminator is discarded at inference. The adversarial head is retained as a lightweight, secondary regularizer for source-condition-invariant features, not as a domain-adaptation module and not as a standalone contribution; the cross-condition experiments in Section 5.6 are reported primarily to characterize the limits of this regularizer rather than to claim deployment-ready domain transfer.

### *3.8. Regression head*

The fused vector is decoded by LayerNorm → Dropout(0.2) → Linear(128→64) → GELU → Dropout(0.1) → Linear(64→1). The head contributes 8,577 parameters.

### *3.9. Monotonic post-processing: offline and causal*

Per-bearing PAVA (offline). At evaluation, each bearing's complete prediction sequence is projected onto the monotone non-increasing cone by the pool-adjacent-violators algorithm [11] (O(n); 346 $\mu$s per 1,500-window bearing on CPU). Per-bearing application preserves inter-bearing variance. PAVA requires the full sequence and is therefore retrospective: because the pool-adjacent-violators projection couples each window to its neighbors, a corrected value may depend on later windows in the trajectory. This is legitimate for retrospective fleet analysis but invalid for streaming prognosis, which is precisely why we evaluate the strictly causal filters below for online deployment.

Causal filters (online). For real-time deployment we evaluate (i) the real-time running minimum RTRM, $\hat{y}_t \leftarrow \min(\hat{y}_t, \hat{y}_{t-1})$, parameter-free and strictly monotone; and (ii) constrained exponential smoothing CES, an EMA ($\alpha = 0.2$) with a bounded upward step ($\varepsilon = 0.05$), which trades strict monotonicity for accuracy. Both operate strictly on past predictions.

### *3.10. Uncertainty quantification*

At evaluation, 30 stochastic forward passes with active dropout (BatchNorm statistics frozen) yield a predictive mean $\mu_t$ and standard deviation $s_t$ per window [14]. We construct the interval lower edge $\mu_t - 1.645\, s_t$, the lower bound of the conventional two-sided 90% Gaussian interval, and report three metrics against a 90% nominal target: lower-bound coverage (LBC), $P(y_t \geq \mu_t - z s_t)$, a one-sided metric chosen to match the lower-bound maintenance-alarm rule of Section 6.3 rather than the two-sided interval-membership rate conventionally denoted PICP; mean prediction-interval width (MPIW); and Gaussian negative log-likelihood (NLL). Because raw MC-dropout intervals under-cover, a leak-free post-hoc calibration is applied: a single variance multiplier is fitted to reach the 90% nominal target on the held-out training bearings of the two validation folds (never on test data) and applied unchanged to the test intervals (Section 5.7). No retraining is required, and the point predictions are unaffected.

### 3.11. Maintenance-threshold decision metrics

Interval coverage alone does not establish that a prediction is decision-relevant: a rule can be well calibrated in aggregate yet still alarm too early or too late on individual trajectories. We therefore reframe the interval-triggered maintenance rule of Section 6.3 in terms of alarm *timing*. For a normalized-RUL maintenance threshold $\tau$ and a bearing's true trajectory $y_t$, the true alarm time is $t_{\text{true}}(\tau)$ = the first t with $y_t \le \tau$; for a given prediction variant $\hat{y}_t$ (raw, PAVA, RTRM, CES, or the calibrated lower bound $\mu_t - zc^* s_t$), the predicted alarm time is $t_{\text{hat}}(\tau)$ = the first t with $\hat{y}_t \le \tau$, clamped to the trajectory end and flagged as censored if the variant never crosses $\tau$. The alarm timing error is

$$\Delta t(\tau) = t_{\text{hat}}(\tau) - t_{\text{true}}(\tau), \tag{7}$$

with $\Delta t < 0$ an early alarm and $\Delta t > 0$ a late alarm. We additionally flag a late alarm as *severe* if it is censored or if the true RUL at the moment the alarm finally fires has already fallen at or below $\tau/2$, an illustrative severity rule tied to the physical trajectory rather than a fixed window count, not a validated industrial criterion. Decisions are scored with an asymmetric cost,

$$L_{\text{decision}}(\tau) = c_{\text{late}} \cdot \max(0, \Delta t) + c_{\text{early}} \cdot \max(0, -\Delta t), \tag{8}$$

reported at a headline ratio $c_{\text{late}}{:}c_{\text{early}} = 5{:}1$ (late maintenance is usually more safety-critical than early maintenance) with 2:1 and 10:1 as sensitivity settings; these ratios are illustrative, not a universal industrial cost model. Section 6.3 evaluates Eqs. 7–8 at $\tau \in \{0.30, 0.20, 0.10\}$ on the same frozen 20-seed test predictions used throughout Section 5, requiring no retraining.

## 4. Experimental protocol and benchmark design

### 4.1. Datasets

PRONOSTIA (IEEE PHM 2012) [15]. Run-to-failure vibration at 25.6 kHz; one 0.1 s snapshot recorded every 10 s. Each snapshot is an independent, non-overlapping window of 2,560 samples; consecutive windows are 10 s apart and share no raw samples, so the every-10th-window in-bearing validation split of Section 4.2 cannot leak shared raw samples between training and validation, although windows drawn from the same bearing trajectory remain correlated through the shared, slowly-varying degradation state, a weaker dependence that non-overlapping windows alone do not eliminate. Following the cross-condition supervised split used throughout this work, the training set comprises Bearing1_1, Bearing1_2 (Condition 1: 1800 rpm, 4 kN) and Bearing2_1, Bearing2_2 (Condition 2: 1650 rpm, 4.2 kN); the test set comprises Bearing1_3, Bearing1_4, Bearing1_5, Bearing2_3. Exact window counts: training 2,803 / 871 / 911 / 797 (total 5,382); test 2,375 / 1,665 / 2,873 / 1,955 (total 8,868).

XJTU-SY [16]. Fifteen run-to-failure bearings, three conditions (35 Hz/12 kN; 37.5 Hz/11 kN; 40 Hz/10 kN), 25.6 kHz sampling, 1.28 s recording per minute. We cut twelve non-overlapping 2560-sample windows per recording (horizontal channel); windows are retained in chronological order within each bearing and supplied to the PGMC auxiliary pass as contiguous within-bearing chunks (Section 3.6), never shuffled as independent instances for the monotonic penalty. RUL labels are assigned at recording (file) granularity, not per subwindow: all twelve subwindows cut from the same one-minute recording file i receive an identical label $(n_{files} - i)/n_{files}$, matching the general $y_t = (T - t)/T$ formulation of Section 3.1 with t indexing recordings rather than individual subwindows; chronological subwindow order is preserved only for PGMC chunk sampling and is never used to assign distinct labels within a recording. Training: Bearing1_1, Bearing1_2, Bearing2_1, Bearing2_2; in-domain test: the six remaining Condition-1/2 bearings; zero-shot: all five Condition-3 bearings (never seen in any form). Lifetimes span 42–2,538 minutes, making the per-window RUL-fraction task substantially harder than PRONOSTIA's.

CWRU [17] (negative control). Seeded-fault recordings at 12 kHz with pseudo-RUL labels constructed from damage-severity sequences. Used only to test naive cross-dataset transfer (Section 6.4).

RUL labels are normalized per bearing to [0,1] as in Section 3.1. Raw vibration windows are used directly in their recorded units, with no additional per-window or per-bearing amplitude normalization or standardization; residual scale differences across bearings and operating conditions are absorbed by the network's BatchNorm layers rather than by input preprocessing.

### *4.2. Why model selection must be leakage-free*

With only four test bearings, checkpoint selection dominates reported performance. We quantify two common practices on PRONOSTIA (full model, 5 seeds, identical training):

1. Early stopping on an in-bearing validation split (every 10th window of each training bearing; patience 30): validation MAE keeps improving while held-out-bearing MAE degrades; the selected checkpoints score PAVA MAE 0.1134 ± 0.0113, 41% worse than the same architecture under our protocol.

2. Early stopping monitored on the test bearings (a practice that can inadvertently arise when code repositories are optimized directly against the test set, masking true generalization capacity): PAVA MAE 0.0750 indistinguishable from our leakage-free result (0.0789 ± 0.0068), demonstrating that the leak primarily

hides the model-selection problem rather than creating accuracy, and that leakage-free selection can recover nearly all of the apparent performance.

### *4.3. Leakage-free two-stage protocol*

Stage 1: budget selection (validation data only). Candidate budgets B ∈ 5, 10, 20, 40 epochs are each trained to completion (cosine annealing over exactly B epochs; budget-scaled warm-ups) on two bearing-level validation folds (fold A: validate on Bearing1_2; fold B: validate on Bearing2_2), five seeds each, a 2-of-4-fold cross-validation design over the training bearings, not a full leave-one-bearing-out sweep. The two folds represent the two distinct operating conditions in the training set (Bearing1_2 at 1800 rpm, Bearing2_2 at 1650 rpm), providing condition-diverse validation coverage with the available training data; a full 4-fold sweep over all training bearings (adding Bearing1_1 and Bearing2_1 as validation targets) would not increase condition diversity and was not pursued. The budget with the lowest mean final-epoch validation MAE is selected, using validation data exclusively. On PRONOSTIA this yields B* = 5 decisively (0.129 vs. 0.148 at B = 10); the same selection performed independently for the batch-ordering PGMC variant also yields B* = 5, and the identical sweep on XJTU-SY yields B* = 5 (0.163 vs. 0.199). The validation-budget sweep is presented in Figure 3. Section 6.1 reports, strictly as post-hoc diagnostics that never informed these choices, how sensitive the test-set outcome would have been to the budget and regularizer-strength decisions made here.

One-standard-error rule for all other hyperparameters. The budget is the only tuned hyperparameter. All remaining settings (learning rate, $\lambda_{\text{PGMC}} = \lambda_{\text{DANN}} = 0.08$, dropout, architecture widths) are inherited defaults. A sensitivity sweep on the same two validation folds over the regularizer strength $\lambda_{\text{PGMC}} \in$ 0.08, 0.2, 0.5, 1.0, 2.0, 5.0 produced validation MAE in the range 0.125–0.129 for $\lambda \leq 2.0$, all differences within one standard error of the fold noise, rising only at $\lambda = 5$. Following the one-standard-error principle [10], we retain the default rather than selecting a noisy argmin over this small validation-only gain; Section 6.1 reports, strictly as a post-hoc diagnostic that never informed this choice, what selecting the argmin ($\lambda = 2.0$) would have cost on the test set.

Stage 2 fixed-budget training. The final model is trained on all four training bearings for exactly B* epochs with no validation and no early stopping, twenty seeds, and evaluated on the test bearings exactly once.

### *4.4. Training configuration*

AdamW (lr $3{\cdot}10^{-4}$, weight decay $10^{-4}$), batch 32, gradient clipping 1.0, cosine annealing $T_{\max} = \text{B*}$, $\lambda_{\text{PGMC}} = \lambda_{\text{DANN}} = 0.08$ with budget-scaled warm-ups, dropout 0.2, feature dimension 128, FP32. A full five-seed PRONOSTIA training run completes in under two GPU-minutes on an RTX 5070; the entire protocol

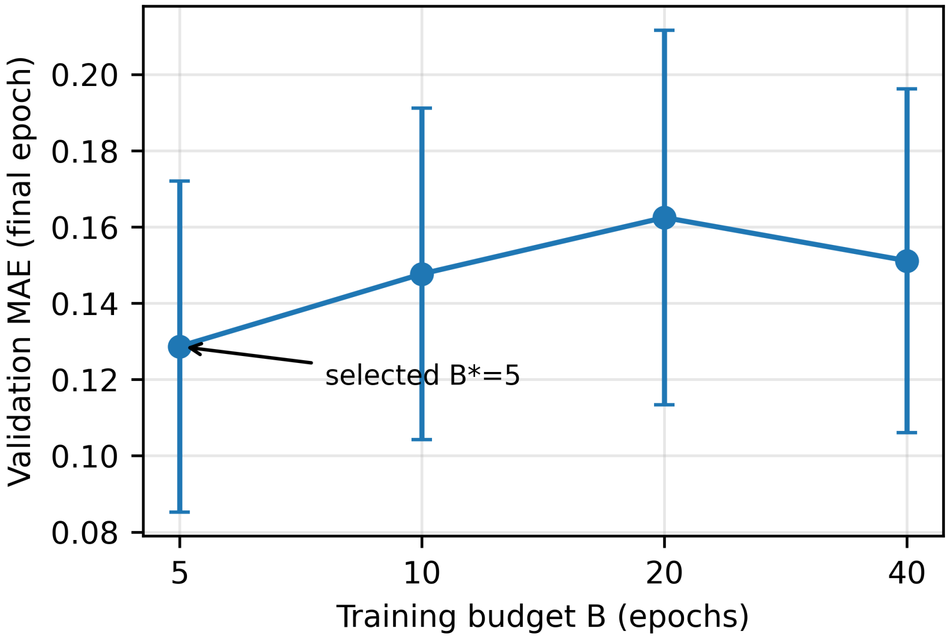


Figure 3: Two-fold bearing-level validation budget sweep on PRONOSTIA: mean final-epoch validation MAE vs. training budget (10 runs per budget). B* = 5 is selected on validation data only.

including budget sweep, ablations, and UQ evaluation is reproducible in a few GPU-hours.

### *4.5. Metrics and statistics*

MAE, RMSE, $R^2$, and strict monotonicity percentage (fraction of consecutive within-bearing pairs with $\hat{y}_{t+1} \leq \hat{y}_t$), pooled over test windows; LBC/MPIW/NLL for UQ. Unless otherwise noted, all results are mean $\pm$ std over 20 fixed seeds (42, 123, 777, 999, 7 plus fifteen pre-specified additional seeds); the sole exception is the XJTU-SY baselines (Table 6), run over 5 seeds paired on a shared subset for tractability, while our own model on both datasets and all PRONOSTIA baselines use the full 20. Because a full training run costs seconds at B* = 5, large seed counts are practical and, as Section 5.3 shows, necessary: five-seed pilot estimates overstated one component effect four-fold. Paired two-sided t-tests over matched seeds with Cohen's d effect sizes are reported throughout. Hardware: AMD Ryzen 5 7500F (6-core, 3.8 GHz base / 5.1 GHz boost, single-threaded sequential inference), NVIDIA RTX 5070 (12 GB VRAM), 16 GB DDR5.

## 5. Results

### 5.1. PRONOSTIA main results

Table 1 summarises per-seed and aggregate performance across 20 fixed seeds.

Table 1: PRONOSTIA test-set results (20 seeds; per-seed values in the released artifacts).

| Output | MAE | RMSE | $R^2$ | Mono% |
|---|---|---|---|---|
| Raw | 0.0897 ± 0.0050 | 0.1234 ± 0.0059 | 0.8169 ± 0.0176 | 50.6 ± 0.6 |
| PAVA | 0.0789 ± 0.0068 | 0.1059 ± 0.0086 | 0.8646 ± 0.0224 | 100.0 ± 0.0 |

Raw monotonicity (50.6%) confirms that pointwise-trained outputs order consecutive predictions no better than chance, even with PGMC active: PGMC is a smooth latent-space regularizer that reshapes feature trajectories to support effective post-hoc isotonic correction, not a hard step-level ordering enforcer. The 50.6% rate is consistent with this regularizer's intended scope rather than a shortfall; the monotonicity guarantee is the exclusive responsibility of the post-processing stage. PAVA improves MAE by 12.0% (0.0897 → 0.0789) and $R^2$ by 4.8 points consistently across seeds, showing that the trained representation, shaped jointly by PGMC and DANN, since neither regularizer reaches significance in isolation (Section 5.3, Table 3), is well-ordered enough for PAVA to recover a clean trajectory from statistically coin-flip raw outputs. A representative predicted trajectory, uncertainty interval, and PAVA-corrected output are shown in Figure 4.

### 5.2. Comparison with baselines on the identical protocol

To put the result in context, we trained five baseline architectures under the identical leakage-free protocol, independent budget sweep, same seeds, same data, plain L1 loss: a 4-block vanilla 1D CNN (channels 32→64→128→256, kernels 15/7/5/3, GAP + linear head); a CNN-BiLSTM (2-layer 1D CNN stem, channels 1→64→128, feeding a 2-layer bidirectional LSTM, hidden size 128); a CNN-LSTM hybrid (3-layer CNN stem, channels 1→64→128→128, feeding a 2-layer bidirectional LSTM, hidden size 128); a 1D transformer encoder (patch-embedded STFT spectrogram input, patch size 8, embedding dimension 256, 4 heads, 4 encoder layers); and a deep 1D ResNet (2-layer strided CNN stem followed by 8 residual blocks, 128 channels).

Throughout Tables 2, 3, and 6, the reported percentage is the comparator's own MAE reduction relative to itself, i.e. (comparator − ours)/comparator, not relative to our value. DSAFLite+PGMC outperforms every baseline (all paired $p < 10^{-4}$ over 20 matched seeds) while tied for the fewest parameters (0.19 M, matching the vanilla 1D CNN). The strongest baseline (CNN-LSTM, 0.80 M parameters) trails by 16.6% ($d = 1.18$); the recurrent models are competitive but 4× larger; the vanilla

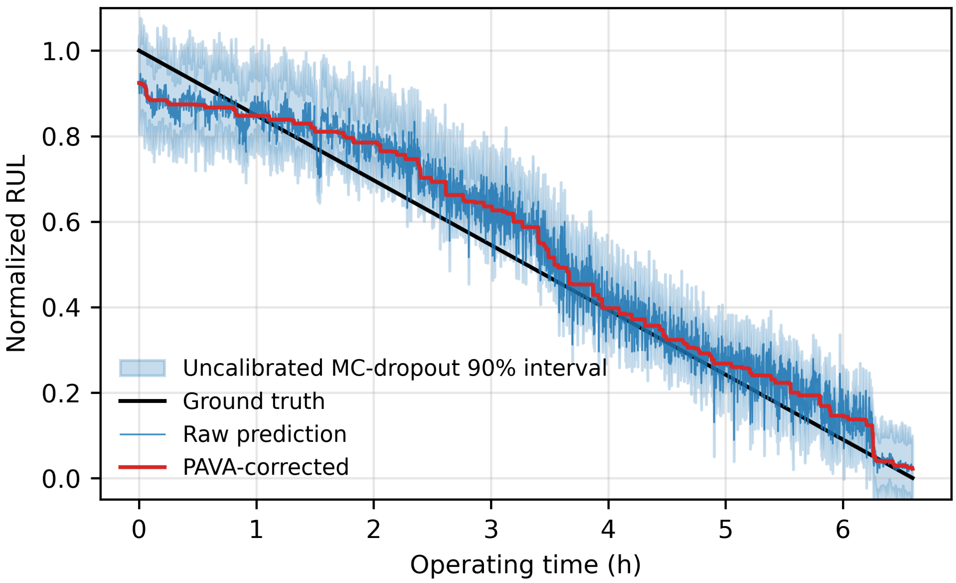


Figure 4: Predicted RUL trajectory with MC-dropout 90% interval, raw and PAVA-corrected predictions (Bearing1_3, seed 42).

Table 2: Baselines on the identical PRONOSTIA protocol (20 seeds; budget K selected per architecture by the same two-fold validation sweep).

| Model | Params | K | Raw MAE | PAVA MAE | Δ vs. ours | p (paired) | d |
|---|---|---|---|---|---|---|---|
| Vanilla 1D CNN | 0.19 M | 40 | 0.1561 ± 0.0437 | 0.1427 ± 0.0436 | +44.7% | $<10^{-4}$ | 1.37 |
| BiLSTM | 0.75 M | 10 | 0.1137 ± 0.0091 | 0.0977 ± 0.0086 | +19.2% | $<10^{-4}$ | 1.41 |
| CNN-LSTM | 0.80 M | 5 | 0.1096 ± 0.0101 | 0.0946 ± 0.0102 | +16.6% | $<10^{-4}$ | 1.18 |
| 1D ResNet | 0.97 M | 10 | 0.1747 ± 0.0505 | 0.1679 ± 0.0498 | +53.0% | $<10^{-4}$ | 1.61 |
| Transformer (STFT patches) | 3.24 M | 10 | 0.2747 ± 0.0377 | 0.2389 ± 0.0400 | +67.0% | $<10^{-4}$ | 3.74 |
| DSAFLite+PGMC (ours) | 0.19 M | 5 | 0.0897 ± 0.0050 | 0.0789 ± 0.0068 | – | – | – |

CNN and 1D ResNet are unstable across seeds ($R^2$ collapses on individual seeds, inflating their variance); and the spectrogram-patch transformer fails outright on a dataset of this size ($R^2 < 0$ on every seed), consistent with the data-hunger of attention models on small run-to-failure benchmarks. All baselines received the identical leakage-free budget-sweep selection, so these gaps are architectural, not artifacts of tuning asymmetry.

Published context. The closest published methods, SC-SAN [19] and HIWSAN [18], report PRONOSTIA MAE of 0.0793 and 0.0788 respectively but under a different protocol (six single-source → single-target transfer tasks with access to unlabeled target-bearing data), so the numbers are not directly comparable to our supervised held-out-bearing setting. We therefore cite these figures as context only, not as a superiority claim: DSAFLite+PGMC reaches a comparable raw-MAE level (0.0789 ± 0.0068) with 13× fewer parameters (0.19 M vs. 2.6 M [18]), without using any target-bearing data during training. HIWSAN's reported 11.32 ms latency [18] is measured on CPU, whereas ours (1.48 ms, Section 5.8) is measured on GPU; because the two figures are not on comparable hardware, we do not report a derived cross-model latency ratio.

### *5.3. Ablation study*

Table 3: Component ablations (identical protocol, B* = 5, 20 seeds, paired tests).

| Variant | Raw MAE | PAVA MAE | PAVA $R^2$ | Δ PAVA MAE | p (paired) | d |
|---|---|---|---|---|---|---|
| Full model | 0.0897 ± 0.0050 | 0.0789 ± 0.0068 | 0.865 ± 0.022 | – | – | – |
| w/o adaptive gate (static 0.5) | 0.0934 ± 0.0068 | 0.0862 ± 0.0076 | 0.840 ± 0.028 | +8.5% | $<10^{-4}$ | 1.17 |
| Temporal stream only | 0.0933 ± 0.0068 | 0.0847 ± 0.0070 | 0.858 ± 0.022 | +6.9% | 0.0006 | 0.93 |
| Spectral stream only | 0.1845 ± 0.0044 | 0.1624 ± 0.0036 | 0.464 ± 0.024 | +51.4% | $<10^{-6}$ | 10.5 |
| w/o PGMC | 0.0911 ± 0.0059 | 0.0810 ± 0.0069 | 0.856 ± 0.029 | +2.5% | 0.28 | 0.25 |
| w/o DANN | 0.0904 ± 0.0048 | 0.0810 ± 0.0051 | 0.852 ± 0.020 | +2.5% | 0.13 | 0.36 |
| w/o PGMC and w/o DANN | 0.0925 ± 0.0062 | 0.0836 ± 0.0066 | 0.839 ± 0.033 | +5.6% | 0.018 | 0.58 |
| PGMC batch-ordering variant | 0.0901 ± 0.0046 | 0.0797 ± 0.0058 | 0.861 ± 0.022 | +1.0% | 0.70 | 0.09 |

Three findings stand out, and we stress that the third is only visible because of the 20-seed design. First, the adaptive gate is the largest single contributor: −8.5% PAVA MAE over static fusion ($p < 10^{-4}$, $d = 1.17$), a substantive reversal of fusion-gate null results reported under leakier protocols; per-bearing mean gate weights vary from 0.33 to 0.47, confirming per-input adaptivity. Second, dual-stream fusion contributes −6.9% over the temporal stream alone ($p < 10^{-3}$), even though the spectral stream is far weaker in isolation: complementarity, not redundancy. Third, the two training-time regularizers behave subtly: PGMC and DANN each contribute ≈2.5% individually, within seed noise (p = 0.28, 0.13), but their joint removal costs a significant 5.6% (p = 0.018, d = 0.58): the two losses partially substitute for each other as regularizers of the shared representation. Five-seed pilot runs had estimated the individual PGMC effect at −10% (d ≈ 1.1); twenty seeds corrected this to −2.5%. We report the corrected decomposition and caution that single-factor ablations with n ≤ 5 seeds on four test bearings can overstate component effects by several-fold. The batch-ordering PGMC relaxation is statistically indistinguishable from the sequential form (d = 0.09), making it a convenient drop-in when sequential batch construction is impractical. The spectral stream's standalone $R^2$ of 0.464 (Table 3) is physically expected: the Hilbert envelope spectrum reveals which defect frequencies are active but discards the timing and phase of individual fault impulses present in the raw vibration, information that carries RUL-correlating waveform morphology. The temporal stream preserves this richer signal. The adaptive gate exploits both: under clean conditions it defaults to the temporally richer stream (mean gate weight $\alpha \approx 0.42$); under noise it shifts toward the demodulation-robust spectral stream ($\alpha$ decreases with SNR), a behaviour consistent with the physical noise-rejection properties of amplitude demodulation.

Warm-up scaling matters. With the original absolute warm-ups (10/20 epochs) and the rigorously selected budget B* = 5, PGMC reaches only half strength and DANN never activates; the apparent ”component contributions” then reduce to seed noise. All ablations above use budget-scaled warm-ups (Section 3.6), which is what makes the comparison meaningful. The component-ablation results are summarized graphically in Figure 5.

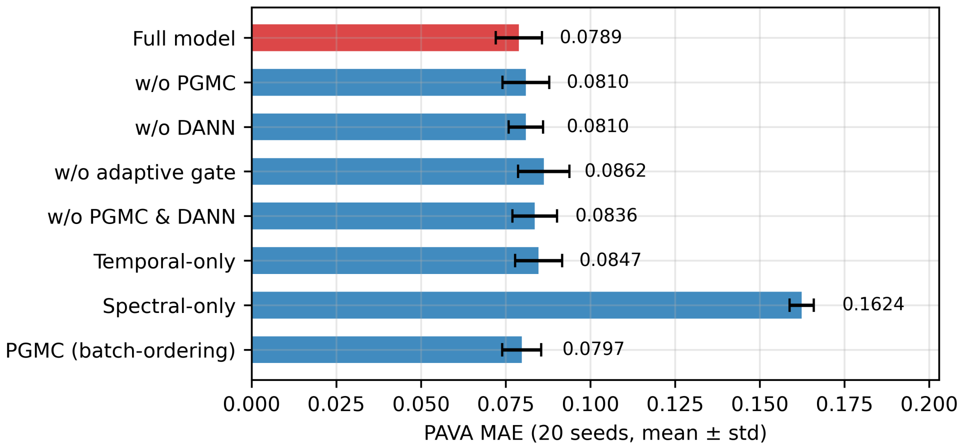


Figure 5: Ablation summary (PAVA MAE, mean ± std).

### 5.4. *Post-processing: offline vs. causal*

PAVA remains optimal when the full sequence is available. Among causal options, CES improves accuracy over raw (−6.2% MAE) but barely improves monotonicity; RTRM guarantees monotonicity at a 27–38% accuracy cost depending

Table 4: Post-processing comparison (PRONOSTIA, 20 seeds, identical raw predictions).

| Method | MAE | RMSE | $R^2$ | Mono% | Causal |
|---|---|---|---|---|---|
| Raw (none) | 0.0897 ± 0.0050 | 0.1234 ± 0.0059 | 0.817 ± 0.018 | 50.6 ± 0.6 | ✓ |
| RTRM | 0.1236 ± 0.0180 | 0.1545 ± 0.0187 | 0.709 ± 0.069 | 100.0 | ✓ |
| CES ($\alpha$=0.2, $\varepsilon$=0.05) | 0.0841 ± 0.0057 | 0.1165 ± 0.0065 | 0.837 ± 0.018 | 50.3 ± 0.6 | ✓ |
| PAVA (per-bearing) | 0.0789 ± 0.0068 | 0.1059 ± 0.0086 | 0.865 ± 0.022 | 100.0 | × |

on prediction drift. For real-time deployment we recommend CES when accuracy dominates and RTRM when guaranteed monotonicity dominates; the choice is application-specific and both are O(1) per step. Figure 6 compares the four post-processing strategies on a representative test bearing.

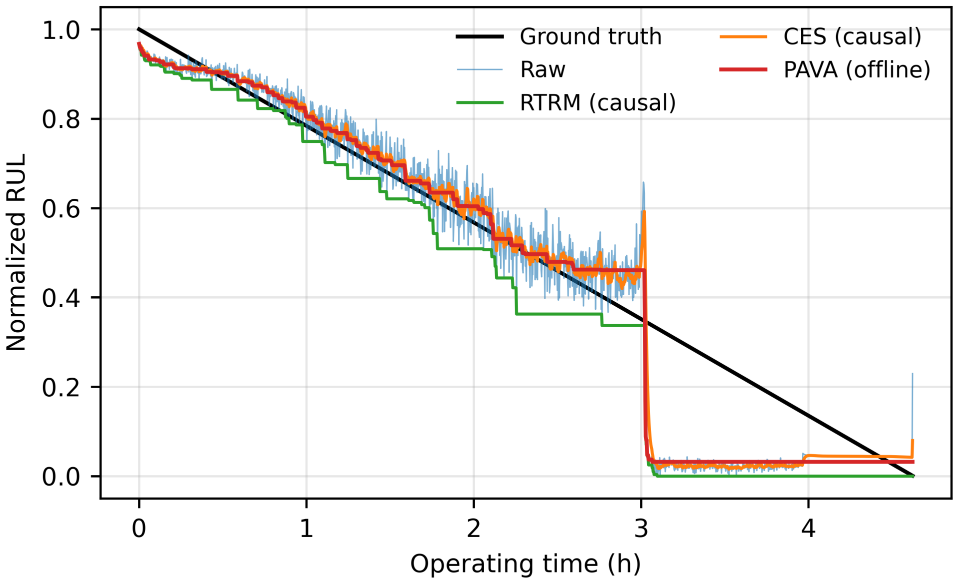


Figure 6: All four post-processing strategies on one test bearing.

### 5.5. Noise robustness and gate behavior

Additive white Gaussian noise was injected into the test windows at controlled SNR (evaluation only; identical checkpoints).

The gate maintains a consistent 4–5% advantage from clean conditions down to 20 dB, narrowing to 3.7% at 10 dB, and reaches parity at 5 dB. Below 0 dB both fusion strategies are unusable ($R^2$ < 0 for both, Table 5) and static fusion degrades more gracefully. The mean gate weight decreases monotonically with SNR (0.418 → 0.386), shifting weight toward the noise-robust envelope-spectrum stream, consistent with the gate's intended role, although the response magnitude

Table 5: Robustness to additive noise (raw MAE and $R^2$, mean over 20 seeds; $R^2$ std not computed for this sweep).

| SNR | Adaptive gate MAE | Static fusion MAE | Gate advantage | Adaptive $R^2$ | Static $R^2$ | Mean $\alpha$ |
|---|---|---|---|---|---|---|
| clean | 0.0897 ± 0.0050 | 0.0934 ± 0.0068 | +4.0% | 0.817 | 0.805 | 0.418 |
| 20 dB | 0.0904 ± 0.0050 | 0.0949 ± 0.0070 | +4.7% | 0.814 | 0.800 | 0.417 |
| 10 dB | 0.1029 ± 0.0077 | 0.1069 ± 0.0094 | +3.7% | 0.762 | 0.753 | 0.415 |
| 5 dB | 0.1462 ± 0.0138 | 0.1475 ± 0.0159 | +0.9% | 0.557 | 0.560 | 0.410 |
| 0 dB | 0.2477 ± 0.0142 | 0.2362 ± 0.0158 | −4.9% | −0.097 | −0.019 | 0.402 |
| −5 dB | 0.3406 ± 0.0217 | 0.3116 ± 0.0223 | −9.3% | −0.966 | −0.708 | 0.386 |

is modest. We report this transparently: the gate is an accuracy feature in nominal-to-moderately-noisy regimes, not a rescue mechanism under extreme corruption. The effects of noise on prediction accuracy and adaptive-gate behavior are shown in Figure 7.

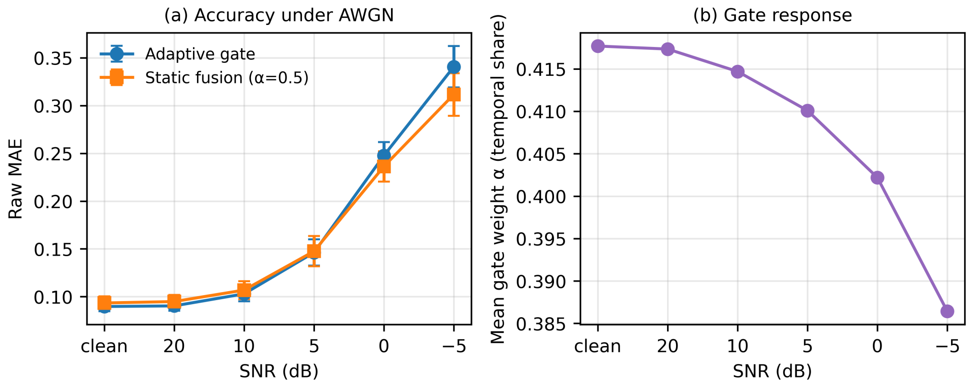


Figure 7: (a) MAE vs. SNR for adaptive vs. static fusion; (b) gate response.

### *5.6. XJTU-SY: second dataset and zero-shot condition transfer*

XJTU-SY is used here as an external stress test of the protocol under a second, harder dataset rather than as evidence of universal cross-dataset transfer. The identical protocol (independent budget sweep, B* = 5; the $\lambda$ sensitivity sweep on XJTU-SY validation independently favors the default $\lambda = 0.08$ outright) was applied to XJTU-SY, with baselines.

XJTU-SY is substantially harder than PRONOSTIA for window-level RUL-fraction estimation: held-out lifetimes span 42–2,538 minutes, so identical degradation states map to very different label values across bearings. DSAFLite+PGMC outperforms the vanilla CNN (p = 0.004); it is numerically better than CNN-LSTM (d = 0.94) though this comparison does not reach conventional significance (p = 0.104), and it is statistically indistinguishable from the strongest baseline (BiLSTM, −3.9%, p = 0.27) while using 4× fewer parameters (0.19 M vs.

Table 6: XJTU-SY results (ours: 20 seeds; baselines: 5 seeds, paired on shared seeds). PAVA MAE is pooled across all test windows (window-count-weighted across the six held-out or five zero-shot bearings), not a simple average of the per-bearing means shown in Fig. 8.

| Model | In-domain raw MAE | In-domain PAVA MAE | Δ vs. ours | p (paired) |
|---|---|---|---|---|
| Vanilla 1D CNN (0.19 M) | 0.2071 ± 0.0089 | 0.1992 ± 0.0104 | +14.1% | 0.004 |
| CNN-LSTM (0.80 M) | 0.2124 ± 0.0208 | 0.1912 ± 0.0226 | +10.5% | 0.104 |
| BiLSTM (0.75 M) | 0.1769 ± 0.0072 | 0.1647 ± 0.0097 | −3.9% | 0.273 |
| DSAFLite+PGMC (0.19 M) | 0.1813 ± 0.0142 | 0.1708 ± 0.0131 ($R^2$ 0.49) | – | – |
| Zero-shot, 5 unseen Cond.-3 bearings | 0.2037 ± 0.0124 | 0.1936 ± 0.0143 ($R^2$ 0.19) | – | – |

0.75 M), we report this tie transparently rather than claiming across-the-board superiority. Measured under the identical batch-1 FP32 GPU protocol of Section 5.8, BiLSTM's own inference latency (1.51 ms median) is statistically indistinguishable from ours (1.48 ms): at this model scale, latency on GPU is dominated by fixed kernel-launch overhead, so the 4× parameter-count advantage does not translate into a measurable latency advantage. The published transfer-task literature reports XJTU-SY MAE of 0.12 [19] under single-target protocols with unlabeled target data; our supervised held-out protocol with six simultaneous test bearings is stricter. Zero-shot transfer to Condition 3 retains a weak predictive signal (PAVA MAE 0.1936, $R^2 = 0.19$), but the gap to in-domain performance ($R^2 = 0.49$) is large enough to render the model unsuitable for deployment-grade prognosis under unseen conditions, an important negative finding about the limits of a source-condition adversarial regularizer that never observes target-condition (Condition-3) data during training. Published transfer-task results reporting substantially higher cross-condition accuracy on these same benchmarks are therefore not directly comparable to the present, stricter source-only setting. We do not read our result as contradicting those studies, but as characterizing what a purely source-only adversarial regularizer can achieve without any access to target-condition data, and we encourage the field to report explicitly whether target-condition data of any kind (even unlabeled) participates in representation learning, model selection, or hyperparameter tuning, since this materially changes what a reported cross-condition number means. Our leak-free protocol characterizes what this complete source-only configuration, including its DANN regularizer, achieves under no-target-data transfer; because we do not report a zero-shot ablation with DANN removed, we cannot isolate DANN's own marginal contribution to (or ceiling on) this result, and we do not claim this finding generalizes to domain-adversarial or transfer-learning methods in general (see Limitations). Operationally, a safety-conscious deployment strategy can compensate for this limit: the running MC-dropout variance provides a proxy for distributional deviation: when the per-window predictive variance exceeds a threshold calibrated on source-condition data (e.g., the 95th percentile of training-bearing variance), the system can flag that predictions are operating outside the reliability envelope and escalate to human review rather than acting autonomously. This variance-based domain-alert mechanism is proposed as a practical safeguard pending a model retrained on the new condition; we have not empirically evaluated whether it correctly flags the Condition-3 zero-shot windows, and it does not require target-domain labels. Per-bearing held-out and zero-shot PAVA errors are reported in Figure 8.

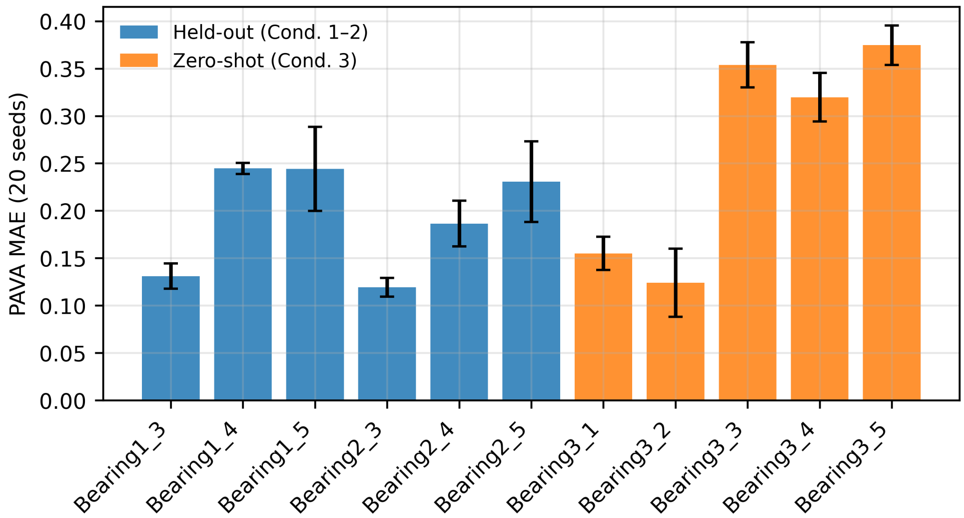


Figure 8: XJTU-SY per-bearing PAVA MAE, held-out and zero-shot. Bars show the unweighted mean PAVA MAE per bearing (across seeds); because bearing lifetimes vary by nearly two orders of magnitude (42–2,538 min), the pooled, window-count-weighted MAE reported in Table 6 is not the arithmetic mean of these bars.

### *5.7. Uncertainty quantification*

As a deployment-oriented secondary analysis to the accuracy and robustness results above, uncalibrated MC-dropout intervals at nominal 90% coverage achieve LBC = 59.9% on PRONOSTIA (MPIW = 0.175): informative but systematically too narrow, which is a well-documented property of MC-dropout with small prediction heads [12, 14]. Post-hoc interval scaling is therefore not an ad hoc workaround but a principled calibration step [23]. We therefore calibrate leak-free: on each of the two validation-fold models (Section 4.3), the held-out training bearing provides per-window ($\mu$, s, y) triples, and the smallest variance multiplier achieving 90% empirical coverage on that bearing is computed. Formally: $c^* = \arg\min_{c\geq 1} |\mathrm{LBC}_{\mathrm{val}}(c) - 0.90|$, where $\mathrm{LBC}_{\mathrm{val}}(c) = \frac{1}{|W_{\mathrm{val}}|} \sum_{t\in W_{\mathrm{val}}} \mathbb{1}[y_t \geq \mu_t - zcs_t]$, $z = 1.645$. The mean across the ten fold × seed runs gives c* = 2.41; this single scalar is applied to all test-set intervals ($s' = c^* \cdot s$). Calibrated intervals achieve LBC = 89.1% at nominal

90% with MPIW = 0.423 and NLL = −0.62. No test data participates in calibration. The calibrated interval (mean width 0.423 normalized RUL, a half-width of ≈ ±0.21 at 90%) faithfully reflects the true predictive uncertainty of window-level RUL estimation on four test bearings wide, but correctly so, and provides a risk-informed input for maintenance-threshold decisions (Section 6.3), subject to the health-stage and per-bearing caveats below. A health-stage breakdown of calibrated coverage exposes the limits of this global scalar: at the early stage ($y > 0.50$) coverage is near-perfect (99.8% pooled); at the mid stage ($0.20 < y \leq 0.50$) it is adequate (91.2% pooled); but at the critical end-of-life stage ($y \leq 0.20$) coverage falls to 71.0% pooled and to only 21.0% for Bearing2_3, the same bearing with 97.0% point-estimate missed detection. This is the quantitative signature of homoscedastic miscalibration: the global $c^* = 2.41$, fitted on training bearings with more gradual final-phase degradation, cannot expand intervals fast enough for Bearing2_3's rapid end-of-life collapse. Three of four test bearings maintain critical-stage coverage above 75%; Bearing2_3 is structurally anomalous. Health-stage binned calibration fitting separate $c^*$ values for each RUL stage would substantially improve coverage uniformity and is the primary planned extension of this work.

### *5.8. Computational efficiency*

Table 7: Computational profile (measured, RTX 5070).

| Metric | Value |
|---|---|
| Inference parameters | 192,130 (0.19 M) |
| Training-only parameters (DANN head) | 8,321 |
| FP32 latency, batch 1 (median / p95) | 1.48 / 1.65 ms |
| Real-time factor (100 ms window) | 68× |
| Throughput, batch 256 | 118,000 windows/s |
| Peak VRAM, batch 1 | 11 MB |
| PAVA per bearing (1,500 windows, CPU) | 346 $\mu$s |
| Full 5-seed training ($B^*$=5) | < 2 GPU-minutes |
| BiLSTM baseline FP32 latency, batch 1 (median), for reference | 1.51 ms |

The model trains end-to-end in seconds per seed; its small footprint (192,130 parameters, 11 MB peak VRAM) suggests suitability for embedded-class deployment, though we measured latency only on the GPU/CPU hardware of Section 4.5 and did not benchmark on an embedded device. Compared with HIWSAN (2.6 M parameters, 33.95 MB, 11.32 ms CPU [18]), the inference graph is 13× smaller; because HIWSAN's figure is CPU-only and ours is GPU-only, this size comparison should not be read as an implicit latency comparison. Measured on the same

GPU under the same protocol, the in-house BiLSTM baseline (0.75 M parameters, Section 5.6) shows no measurable latency advantage for our smaller model (1.48 vs. 1.51 ms): at this parameter scale, GPU inference time is dominated by fixed kernel-launch overhead rather than FLOPs, so parameter-count reductions this small do not translate into measurable latency gains. All latency figures were measured under native PyTorch FP32 (torch.$no_{\text{grad}}$()) without ONNX Runtime, TorchScript, or any compilation acceleration; further speed-up through runtime optimization is straightforward and outside the scope of this evaluation.

## 6. Discussion

### *6.1. What leakage-free evaluation changes*

Three conclusions from earlier evaluation practice reversed under the leakage-free protocol. (i) The adaptive gate's contribution changed from null to −8.5% MAE: in-bearing early stopping had selected over-trained checkpoints whose seed noise swamped the gate effect. (ii) The optimal training budget collapsed from 40–120 epochs (implied by in-bearing validation) to 5 epochs: cross-bearing generalization peaks far earlier than within-bearing fit. (iii) Component warm-ups designed for long schedules silently disabled components at rigorously selected budgets. We suggest that PHM studies report their model-selection channel (what data chose the checkpoint and hyperparameters) as routinely as they report metrics; with four test bearings, that channel determines the result.

Cautionary post-hoc observations on hyperparameter selection. After the protocol was frozen, we checked, purely as a diagnostic that played no role in selecting the final configuration, what two alternative choices would have produced on the test set. First, switching the training budget from the selected $B^* = 5$ to $B^* = 10$ produces test PAVA MAE within one standard error of the selected budget (< 2% relative change): the 2-fold validation choice of $B^*$ is robust to this alternative. Second, selecting the regularizer strength $\lambda$ by the argmin of the two-fold validation sweep (an apparent 3% validation gain at $\lambda = 2.0$, within one standard error of fold noise) would have produced a materially worse outcome: test PAVA MAE degrades from 0.0789 to 0.0930: an 18% loss from a "validated" choice. Cross-bearing validation with two folds is an unbiased but high-variance estimator; argmin selection over it is over-tuning by another name. The one-standard-error rule (Section 4.3) is a vital statistical safeguard on small PHM benchmarks rather than a pedantic distinction it is the difference between a real and an illusory improvement.

### 6.2. *Monotonicity: soft regularization, post-processing, and their division of labor*

PGMC contributes to accuracy jointly with DANN (−5.6%, $p = 0.018$) but does not raise raw monotonicity above chance at the default strength the squared-hinge penalty reshapes the loss landscape for better isotonic correction, not step-level ordering. Guaranteed monotonicity comes from post-processing: PAVA offline and RTRM online; CES, the accuracy-oriented causal alternative, trades that guarantee for lower error and reaches only 50.3% monotonicity (Table 4). This division of labor is the deployable pattern: PGMC structures the latent trajectory so that PAVA's isotonic projection yields a consistent 12.0% MAE improvement (0.0897 → 0.0789), while the post-processor supplies the hard monotonicity guarantee that PGMC never claimed to supply alone. A raw monotonicity near 50% is consistent with this division of labor rather than a shortcoming, and the right place to audit ordering compliance is the PAVA/RTRM/CES output, not the model logits.

### 6.3. *Deployment-oriented interval calibration, decision-rule sensitivity, and alarm timing*

A deployed prognostic system does not act on the point RUL; it acts on a decision rule [31]. We analyze the canonical interval-triggered rule: maintenance is scheduled when the lower bound of the predictive interval falls at or below a lead-time threshold $\tau$ (the normalized remaining life needed to procure parts and schedule a shutdown), with $z = 1.645$, the conventional two-sided-90% Gaussian half-width used as an uncalibrated starting scale; the realized one-sided coverage is brought to the 90% nominal target by the calibration factor c* (Section 5.7), not by the raw Gaussian quantile alone: $z = 1.645$ is not itself a calibrated one-sided-90% bound, which is precisely why calibration is necessary. We compare three policies on the PRONOSTIA test bearings: a point-estimate rule ($\mu_t \leq \tau$), an uncalibrated-interval rule ($\mu_t - z\, s_t \leq \tau$), and a calibrated-interval rule ($\mu_t - z\, c^*\, s_t \leq \tau$, $c^* = 2.41$ fitted leak-free per Section 5.7). A window should alarm when its true RUL satisfies $y_t \leq \tau$; the false-alarm rate is the fraction of non-alarm windows that nonetheless trigger, and the missed-detection rate is the fraction of alarm windows that fail to trigger. Table 9 disaggregates these rates by individual test bearing (20 seeds per bearing); Table 8 aggregates them as a macro-average at the representative threshold $\tau = 0.20$.

Calibration is not uniformly beneficial in magnitude (Table 9): missed detection on Bearing1_3 collapses from 26.4% to 0.1%, while Bearing2_3 whose trajectory spends almost no time above $\tau = 0.20$ in a way the model can recognize improves from 97.0% to 41.5% but remains high, underscoring that per-unit evaluation, not pooled averages, is the correct basis for deployment decisions. The macro-averaged rates in Table 8 compress this heterogeneity: point-estimate MD

Table 8: Realized error rates of an interval-triggered maintenance rule at threshold $\tau = 0.20$ (PRONOSTIA, 20 seeds). For each seed, the rate is macro-averaged across the 4 test bearings; the table reports the mean and standard deviation of that macro-average across the 20 seeds; this std reflects seed-to-seed variability, not bearing-to-bearing variability, which is substantially larger (Table 9).

| Decision policy | False-alarm rate (%) | Missed-detection rate (%) | LBC, empirical (90% nominal target) |
|---|---|---|---|
| Point estimate ($\mu \leq \tau$) | 3.1 ± 1.4 | 43.3 ± 3.4 | – |
| Uncalibrated lower bound (z = 1.645) | 4.7 ± 1.2 | 35.1 ± 3.8 | 59.9% |
| Calibrated lower bound (c* = 2.41) | 15.2 ± 5.2 | 19.1 ± 6.5 | 89.1% |

Table 9: Per-bearing realized error rates of the interval-triggered maintenance rule at $\tau = 0.20$ (PRONOSTIA, 20 seeds per bearing; mean ± std). Pt = point-estimate policy ($\mu_t \leq \tau$); Unc = uncalibrated lower-bound policy ($\mu_t - \text{z}{\cdot}s_t \leq \tau$, z = 1.645); Cal = calibrated lower-bound policy targeting 90% nominal coverage ($\mu_t - \text{z}{\cdot}\text{c*}{\cdot}s_t \leq \tau$, c* = 2.41). MD = missed-detection rate (fraction of critical windows with true RUL $\leq \tau$ that did not trigger); FA = false-alarm rate (fraction of safe windows with true RUL $> \tau$ that triggered). Bearing2_3 dominates the pooled result: its degradation trajectory spends almost no time above $\tau = 0.20$ in a way the model can recognize, so point-estimate missed detection reaches 97%. Calibrated intervals substantially reduce missed detections for Bearing1_3 (26% → 0.1%) and Bearing2_3 (97% → 42%), at the cost of more false alarms, the intended reliability trade-off.

| Test bearing | Pt MD (%) | Pt FA (%) | Unc MD (%) | Unc FA (%) | Cal MD (%) | Cal FA (%) |
|---|---|---|---|---|---|---|
| Bearing1_3 | 26.4 ± 5.7 | 1.0 ± 0.5 | 5.8 ± 2.5 | 4.9 ± 1.6 | 0.1 ± 0.3 | 15.9 ± 2.5 |
| Bearing1_4 | 22.2 ± 11.6 | 11.3 ± 5.8 | 19.2 ± 12.2 | 12.6 ± 5.3 | 11.8 ± 10.2 | 17.6 ± 3.9 |
| Bearing1_5 | 27.4 ± 2.1 | 0.0 ± 0.0 | 26.1 ± 0.1 | 0.3 ± 0.6 | 22.9 ± 3.2 | 13.9 ± 10.2 |
| Bearing2_3 | 97.0 ± 1.1 | 0.0 ± 0.1 | 89.1 ± 9.1 | 0.8 ± 1.3 | 41.5 ± 22.0 | 13.5 ± 10.3 |

= 43.3% ± 3.4%, uncalibrated MD = 35.1% ± 3.8%, calibrated MD = 19.1% ± 6.5%.

Restoring coverage from 59.9% to 89.1% with the leak-free scale c* roughly halves the missed-detection rate (35.1% → 19.1%) in exchange for more false alarms (4.7% → 15.2%), and the calibrated policy attains the lowest missed-detection rate at every threshold $\tau \in 0.10, 0.15, 0.20, 0.30$ (Fig. 9). The planner's practical input is the interval, not the point error: a normalized MAE of 0.08 corresponds to a mean RUL error of roughly ±34 minutes on a 7-hour accelerated-degradation life, and the calibrated 90% interval (mean width ≈ 0.42 normalized RUL) is the realistic budget within which $\tau$ must be set, a result best interpreted jointly with the operating costs of a given asset class rather than as a universal threshold recommendation. The residual 41.5% calibrated missed-detection rate on Bearing2_3-type rapid-collapse trajectories indicates that global interval calibration alone is insufficient there; a health-stage binned calibration strategy (Section 5.7) combined with the variance-based domain-alert mechanism (Section 5.6) would likely improve this further.

We report these as relative error-rate trade-offs rather than absolute costs; Table 9 provides the per-bearing breakdown, and no asset-specific cost ratio is assumed.

The classification-style view above asks whether a window triggers, but not when. In a real deployment, once a rule first triggers, maintenance is scheduled (there is no un-triggering), so the operationally relevant quantity is the timing of that first crossing, not a per-window trigger rate. We therefore recompute the same frozen 20-seed PRONOSTIA test predictions under the alarm-timing metrics of Section 3.11 for five prediction variants: the raw point mean, PAVA (offline), RTRM and CES (causal), and the calibrated lower bound applied directly to the raw point mean $\mu_t$ with no additional trajectory smoothing.

Table 10: Maintenance-threshold alarm-timing and decision-loss analysis at $\tau = 0.20$ (PRONOSTIA, 20 seeds, macro-averaged over the 4 test bearings per seed). $\overline{\Delta t}$ is the mean signed alarm-timing error in windows (10 s each; PRONOSTIA snapshot cadence, Section 4.1); Late/Severe-late/Early are the fractions of bearing-seed trajectories in each category (Section 3.11); $L_{5:1}$ is the mean cost-weighted decision loss at the headline 5:1 late:early cost ratio.

| Variant | $\overline{\Delta t}$ (windows) | Late (%) | Severe-late (%) | Early (%) | $L_{5:1}$ |
|---|---|---|---|---|---|
| Raw | −42.5 | 40.0 | 10.0 | 60.0 | 525.5 |
| PAVA (offline) | +143.8 | 87.5 | 25.0 | 12.5 | 873.3 |
| RTRM (causal) | −42.5 | 40.0 | 10.0 | 60.0 | 525.5 |
| CES (causal) | +79.3 | 66.2 | 25.0 | 33.8 | 803.0 |
| Calibrated lower bound | −834.1 | 0.0 | 0.0 | 100.0 | 834.1 |

Table 10 exposes two findings not visible in the classification view. First,

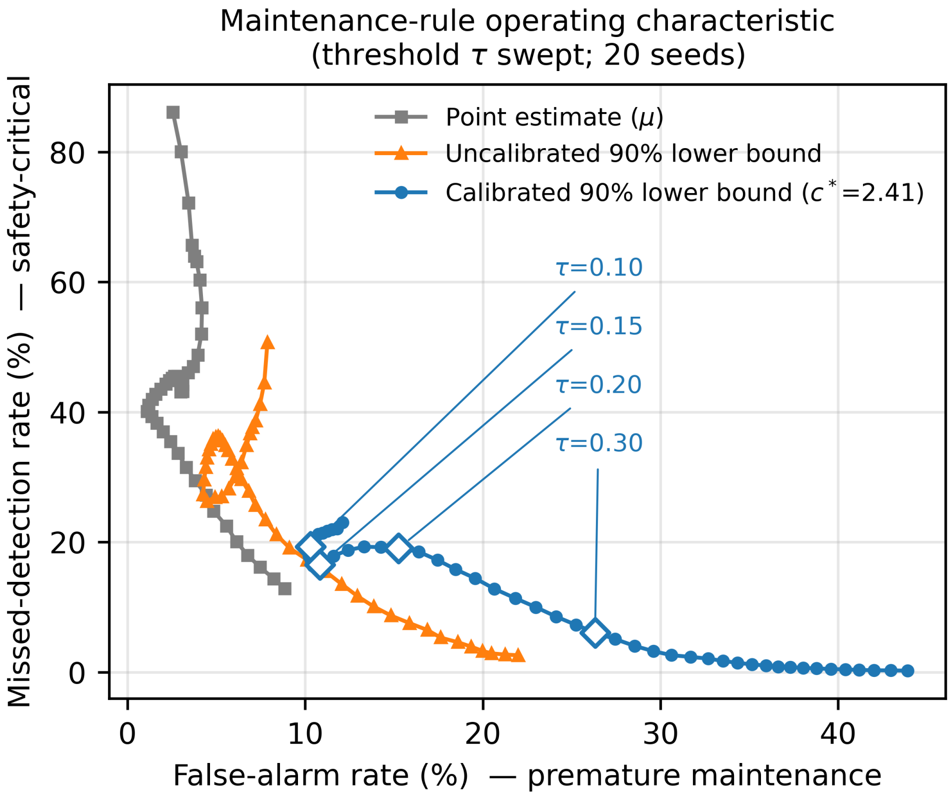


Figure 9: Operating characteristic of the interval-triggered maintenance rule as the threshold $\tau$ is swept: missed-detection rate versus false-alarm rate for the point-estimate, uncalibrated-interval, and calibrated-interval policies (PRONOSTIA, 20 seeds). On average across bearings, the calibrated policy achieves the lowest missed-detection rate in this low-missed-detection regime, though per-bearing performance varies substantially (Table 9).

RTRM is mathematically identical to Raw for alarm timing at every threshold tested: because $\hat{y}_t^{\mathrm{RTRM}} = \min(\hat{y}_t, \hat{y}_{t-1}^{\mathrm{RTRM}})$ is a running minimum, it can cross $\tau$ no earlier or later than the first moment the raw signal itself does: a monotone envelope cannot fall below a threshold before the signal it envelopes does. RTRM's zero-parameter monotonicity guarantee (Table 4) is therefore free with respect to alarm timing, but it is not an improvement either; its value is trajectory-level consistency for downstream visualization and trend interpretation, not earlier or later detection. Second, and more consequentially, the estimator with the best point accuracy is not the one with the best decision outcome. PAVA improves PRONOSTIA MAE by 12.0% over raw (Section 5.1) precisely because its retrospective, whole-trajectory smoothing removes early, noise-driven dips in the raw prediction, but those same dips are exactly what causes Raw to alarm early. Removing them makes PAVA late 87.5% of the time (severe-late 25.0%) at $\tau = 0.20$, versus 40.0% (severe-late 10.0%) for Raw, and under the headline 5:1 late-averse cost, PAVA's decision loss (873.3) is 66% higher than Raw's (525.5). Raw/RTRM achieves the lowest 5:1 decision loss at all three thresholds tested ($\tau \in \{0.30, 0.20, 0.10\}$, full table in the released artifacts), while PAVA is the highest at $\tau \in \{0.30, 0.20\}$ and second-highest (behind the calibrated lower bound) at $\tau = 0.10$.

The ranking is sensitive to the assumed cost ratio, exactly as the sensitivity setting of Section 3.11 anticipates: at a mild 2:1 ratio, Raw/RTRM remains cheapest at $\tau \in \{0.30, 0.20\}$, but PAVA becomes marginally cheaper at $\tau = 0.10$ (295.1 vs. 317.8); at a strongly late-averse 10:1 ratio, the calibrated lower bound, despite its very early mean alarm ($\overline{\Delta t} \approx -139$ minutes at $\tau = 0.20$), becomes cheapest at $\tau = 0.10$ (682.9 vs. 803.3 for Raw), because a rule that can never structurally be late pays no penalty as $c_{\mathrm{late}}$ grows, while Raw's residual late rate is increasingly punished. No single variant is decision-optimal across all cost assumptions; a sensitivity sweep over the cost ratio, not a single assumed value, is the appropriate basis for choosing a deployment rule.

The calibrated lower bound in Table 10 is applied directly to the noisy raw point mean $\mu_t$ with no trajectory smoothing; because the calibrated interval half-width ($\approx$0.21, Section 5.7) is roughly constant across the trajectory while $\mu_t$ fluctuates, a single early transient dip is sufficient to trigger the rule permanently, well before the true crossing (100% early rate at every threshold tested). This is not a failure of calibration (coverage is verified separately in Section 5.7) but a caveat about combining an uncalibrated point trajectory with a calibrated margin: a deployment using the lower-bound trigger should apply it to a filtered mean (e.g., CES) rather than the raw mean, which we did not evaluate here and flag as a direction for future work (Section 6.5). A representative comparison of threshold-crossing behavior is shown in Figure 10.

Table 11 summarizes the reliability threats addressed in this paper and the corresponding control, tying together the leakage, monotonicity, uncertainty, and

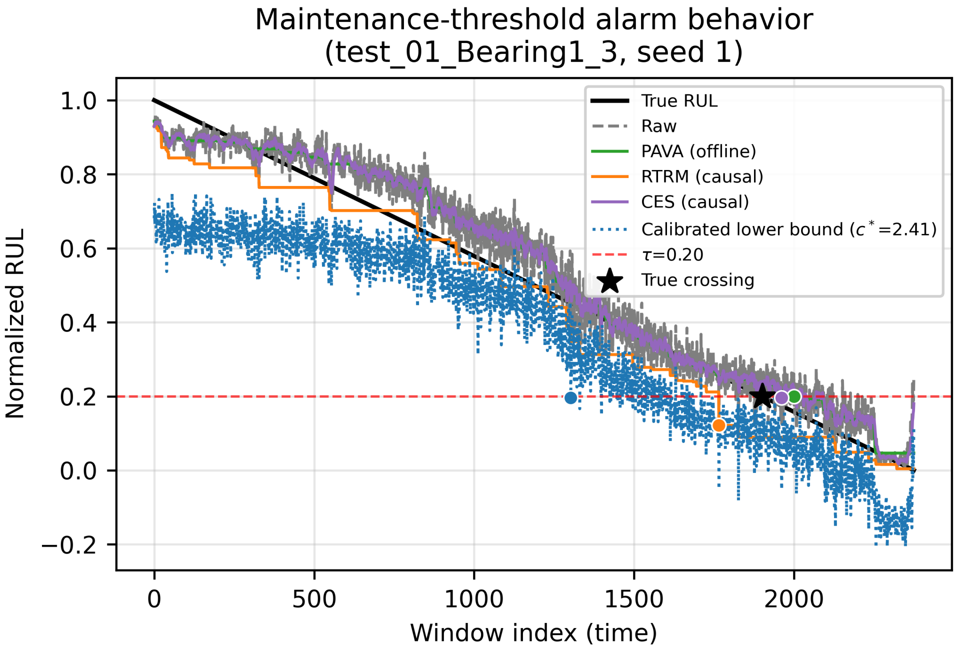


Figure 10: Maintenance-threshold alarm behavior for one representative test bearing and seed at $\tau = 0.20$: true RUL, raw prediction, PAVA, RTRM, CES, and the calibrated lower bound, with the true and predicted first-crossing points marked. RTRM overlaps Raw's crossing exactly, illustrating the timing tautology discussed above.

decision-timing analyses above.

Table 11: Reliability threats addressed in this paper and the corresponding methodological control.

| Reliability threat | Consequence if uncontrolled | Control used in this paper |
|---|---|---|
| Test-bearing model-selection leakage | Optimistic, non-generalizing performance claims | Bearing-level two-fold training-only budget selection (Sections 4.2– 4.3) |
| Non-monotonic RUL trajectory | Inconsistent, physically implausible degradation signal | PGMC training regularizer + PAVA/RTRM/CES post-processing (Sections 3.6, 3.9) |
| Uncalibrated predictive interval | False confidence in a maintenance threshold | Leak-free variance calibration on validation-fold residuals (Sections 3.10, 5.7) |
| Point-accuracy-optimal ≠ decision-optimal | Best-MAE method can be the worst-timed alarm | Alarm-timing and cost-weighted decision-loss analysis (Sections 3.11, 6.3) |
| Operating-regime shift | Unreliable extrapolation beyond validated conditions | Separate in-domain / zero-shot XJTU-SY evaluation (Section 5.6) |

### 6.4. *Negative result: naive cross-dataset transfer fails*

PRONOSTIA-trained checkpoints evaluated zero-shot on CWRU pseudo-RUL labels yield MAE 0.529 ± 0.042 and $R^2 = -3.24$: worse than predicting the mean. The causes are structurally different sampling rate (12 vs. 25.6 kHz, so a 2560-sample window spans 213 ms), seeded faults vs. natural degradation, and pseudo-RUL labels constructed from damage-severity orderings rather than time-to-failure. We report this to correct an error class we found easy to commit: a prior internal evaluation that trained on CWRU with a random window split (placing windows from the same recordings in train and test) produced $R^2 = 0.977$ and could be mistaken for zero-shot generalization. Cross-dataset claims should be validated with strictly disjoint data and native run-to-failure labels, as in our XJTU-SY Condition-3 protocol.

### 6.5. *Limitations*

#### 6.5.1. *Dataset scale and unit-count limitations*

Twenty seeds give adequate power against seed variance, but four to six test bearings remain few; bearing-level conclusions (e.g., per-bearing difficulty) are anecdotal. The alarm-timing analysis of Section 6.3 is computed on these same four PRONOSTIA test bearings, so its per-variant rankings share this small-test-unit caveat.

#### 6.5.2. *Operating-regime and transfer limitations*

XJTU-SY absolute accuracy is constrained because window-level RUL-fraction estimation without temporal context is partly ill-posed under heterogeneous lifetimes; sequence-level context or health-stage detection are promising extensions. Zero-shot condition transfer retains only weak signal; condition-invariant training (DANN) alone is insufficient for deployment-grade transfer, and Condition 3 bearings appear nowhere in training or adversarial passes: the gradient-reversal discriminator sees only Conditions 1 and 2 throughout and is discarded at inference. Because we do not report a Condition-3 zero-shot ablation with DANN removed, the observed zero-shot ceiling ($R^2 = 0.19$, Section 5.6) reflects the complete source-only configuration and cannot be attributed specifically to the DANN regularizer versus the absence of target-condition data in general. Both datasets involve rolling-element fatigue only; gears, compound faults, and variable-speed regimes remain untested, and the envelope-spectrum branch is evaluated under the available near-constant-speed operating regimes only: variable-speed operation would require order tracking or speed-normalized spectral representations before the same demodulation front-end could be applied unchanged.

#### 6.5.3. *Calibration and end-of-life uncertainty limitations*

UQ calibration uses two held-out training bearings; the scale factor's own variance ($\pm 0.9$ across folds) would shrink with more calibration units, and conformal alternatives deserve study. The global scalar $c^* = 2.41$ further assumes homoscedastic miscalibration across the bearing life-cycle, a principled first-order correction suited to operational constraints but unable to capture the heteroscedastic rise in predictive uncertainty near end-of-life (Section 5.7: critical-stage LBC falls to 71.0% pooled and to 21.0% for Bearing2_3); time-varying or conformal prediction intervals [40, 41] are the natural next step. The variance-based domain-alert mechanism proposed in Section 5.6 is a design proposal, not an empirically validated one: we have not tested whether its predictive-variance threshold correctly discriminates in-distribution from Condition-3 zero-shot windows.

#### 6.5.4. *Decision-cost and threshold-model assumptions*

The 5:1 (and 2:1/10:1 sensitivity) late-vs-early cost ratios used in the alarm-timing analysis of Section 6.3 are illustrative choices, not a validated industrial cost model, and the ranking of variants is shown to change with the assumed ratio. The calibrated lower-bound decision trigger evaluated there is applied to the raw, unsmoothed predictive mean; its extremely early alarm behavior is therefore partly an artifact of combining a calibrated interval margin with an uncalibrated point trajectory, and a filtered mean (e.g., CES) under the same trigger was not evaluated.

### *6.5.5. Deployment and online-filter implementation*

CES filter initialization: at t = 0, the constrained exponential smoothing filter initializes directly to the first raw prediction (smoothed[0] = predictions[0]), imposing no warm-up lag. The $\varepsilon = 0.05$ upward-step bound limits per-window RUL increase and prevents artificial inflation of early-stage estimates, a deliberately safety-conservative design: even if the model's first-window output is conservative, corrections can only grow at most 0.05 normalized RUL per step.

### *6.5.6. Architecture and label-formulation scope*

The term 'physics-guided' refers to two architectural elements: (a) the Hilbert-envelope spectral stream, which implements engineering-standard amplitude demodulation to expose bearing defect frequencies, and (b) the PGMC loss, which encodes the expected degradation direction as a physics-consistent training prior. The PGMC is not a hard output-ordering enforcer; increasing $\lambda$ beyond the validation-selected default degrades test MAE by 18%, so the division of labor between physics-informed training and post-hoc monotone projection is an intentional design choice, not a performance shortfall. PGMC chunk batches ($N_c = 32$) are strictly within-bearing and chronologically ordered: no chunk crosses a bearing boundary or a training/validation fold split. The RUL label $y_t = (T - t)/T$ is a strict linear ramp over $t \in 1, \ldots, T$ (Section 3.1), from t = 1 (y = (T−1)/T, just under 1.0) to t = T (y = 0 exactly), identical for both PRONOSTIA and XJTU-SY; no piecewise stage function is used.

## 7. Conclusion

This paper developed a reliability-oriented framework for vibration-based bearing RUL prediction in which model selection, monotonicity, uncertainty calibration, and alarm-timing decision behavior are treated as central parts of the prognostic claim, not as after-the-fact checks on a point-accuracy result. The compact DSAFLite+PGMC estimator instantiates this framework: it combines raw-vibration temporal evidence with Hilbert-envelope spectral evidence through an adaptive scalar gate and is evaluated under a bearing-level protocol that prevents held-out bearings from influencing training-budget selection. On PRONOSTIA, the 0.19 M-parameter model achieved raw MAE/$R^2$ of 0.0897 ± 0.0050/0.8169 ± 0.0176 and PAVA-smoothed MAE/$R^2$ of 0.0789 ± 0.0068/0.8646 ± 0.0224 over 20 seeds, outperforming five baselines trained under the identical leakage-free protocol. Ablation results showed that adaptive fusion (8.5% PAVA MAE reduction, $p < 10^{-4}$) and dual-stream learning (6.9%, $p < 10^{-3}$) are the primary significant architectural contributors; PGMC and DANN are soft auxiliary priors whose joint removal is significant (5.6%, p = 0.018) but whose individual effects are not (Section 5.3), consistent with their role as training-time regularizers rather than accuracy-driving components in their own right. Monotonicity experiments clarified the division of labor between the soft PGMC regularizer, offline PAVA correction, and causal RTRM/CES filters suitable for online deployment. Leak-free uncertainty calibration raised interval coverage from 59.9% to 89.1% at nominal 90%, roughly halving the missed-detection rate of an interval-triggered maintenance rule. Reframing that rule in terms of alarm timing rather than point classification showed that the estimator favored by point-accuracy metrics is not always the one favored by the decision rule: PAVA, despite the best MAE, is late 66–88% of the time and incurs the highest or near-highest cost-weighted decision loss at every threshold tested, while the ranking itself is sensitive to the assumed late-vs-early cost ratio. The noise study showed that the fusion gate increasingly favors the spectral stream as SNR decreases. The XJTU-SY experiments confirmed in-domain usefulness but also showed that zero-shot cross-condition transfer remains unresolved. Overall, the results support compact, signal-informed neural prognostics for reliability-centered maintenance only within the validated operating regimes, and only when model selection, monotonicity, uncertainty calibration, and the decision rule itself are evaluated jointly rather than through point-accuracy metrics alone. Future work should extend the framework to variable-speed operation, compound faults, field data, health-stage-conditional calibration, a filtered (rather than raw) mean under the lower-bound decision trigger, and conformal or distribution-shift-aware uncertainty calibration.

## CRediT authorship contribution statement

Author contributions are reported using the CRediT (Contributor Roles Taxonomy) framework in Table 12.

Table 12: CRediT authorship contribution statement.

| CRediT role | S.M.R.H. Rifat | T. Zhang | A. Labani | B.M.D. Eddine | S.M.A. Ousmane |
|---|---|---|---|---|---|
| Conceptualization | ✓ | | | | |
| Methodology | ✓ | | | | |
| Software | ✓ | | | | |
| Validation | | | ✓ | | ✓ |
| Investigation | ✓ | | | | |
| Resources | | ✓ | | | |
| Data curation | | | ✓ | | |
| Writing – original draft | ✓ | | | | |
| Writing – review & editing | | ✓ | ✓ | ✓ | |
| Visualization | | | ✓ | | |
| Supervision | | ✓ | | | |
| Project administration | | ✓ | | | |

## Code and data availability

The complete training, evaluation, and figure-generation code, configuration files, trained model checkpoints, and result artifacts for the headline PRONOSTIA results are publicly available at https://github.com/Rifatsarkar3/Dual-Stream-Vibration. PRONOSTIA, XJTU-SY, and CWRU are public third-party datasets and are not redistributed in this repository; see the repository README for links to their original sources.

## Declaration of competing interest

The authors declare no competing interests.


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## Declaration of generative AI use

During preparation of this work, the authors used Claude (Anthropic) for language polishing and structural editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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